

Emergency Hospital Resource & Bed Management System

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Project Task–1: Business Requirements & Project Scope

1. Introduction & Motivation

During medical emergencies such as road accidents, pandemics, natural disasters, or mass casualty events, hospitals experience extreme pressure in managing limited resources. Critical resources such as ICU beds, emergency wards, ventilators, oxygen cylinders, and medical staff must be allocated efficiently and in real time. In many cases, lack of a centralized and reliable system leads to delays in patient admission, improper utilization of resources, and increased risk to patient lives.

The objective of this project is to design a database-backed system that helps hospitals manage patient admissions, bed availability, medical resources, and doctor assignments efficiently during emergency situations. The system focuses on real-time visibility, consistency of data, and fast decision-making.

2. System Overview

The **Emergency Hospital Resource & Bed Management System** is a database application that maintains information related to hospitals, wards, beds, patients, medical resources, inventory levels, and doctors. The system supports emergency-based prioritization of patients and enables hospital administrators and medical staff to monitor and allocate resources effectively.

The system ensures that:

- Patients are admitted only when appropriate beds are available
- Critical patients are prioritized
- Hospital resources are monitored continuously
- Shortages of essential resources are detected in advance

3. Scope of the System

3.1 Functional Scope (What the system will do)

The system will support the following functionalities:

- Maintain patient details including emergency level (Critical / Serious / Stable)
- Store information about hospitals, wards, and beds
- Track real-time availability of beds across wards
- Manage patient admissions and discharges
- Monitor hospital inventory of critical resources such as oxygen cylinders and ventilators
- Maintain threshold levels for resources and identify shortages
- Store doctor information and their availability status
- Associate doctors with patients during treatment
- Support concurrent access by multiple hospital staff during emergencies

3.2 Out of Scope (What the system will not do)

- No billing, insurance, or payment processing
- No integration with external medical devices or sensors
- No real-time ambulance tracking or GPS services

4. Stakeholders / Users

The primary stakeholders and users of the system include:

- Hospital administrators
- Doctors and nurses
- Emergency ward staff
- Health authorities (for monitoring hospital readiness)

5. Key Data Entities (High-Level)

The system will manage the following key entities:

- Patient
- Hospital
- Ward
- Bed
- Admission
- Resource
- Inventory
- Doctor

These entities are interrelated and will be modeled using appropriate relationships and constraints in later stages of the project.

6. Business Rules & Constraints

The following business rules govern the system:

- A bed can be assigned to only one active patient admission at a time
 - A patient can have multiple admissions over time
 - ICU and emergency beds are prioritized for critical patients
 - A bed becomes available automatically after patient discharge
 - Resource inventory quantity cannot be negative
 - Alerts are generated when resource quantity falls below the threshold level
 - A doctor can attend multiple patients simultaneously
 - Patient admission is allowed only if suitable resources are available
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7. Need for a Database Management System

A DBMS is essential for this system due to:

- Concurrent access by multiple users during emergencies
 - Maintaining consistency and integrity of critical hospital data
 - Enforcing constraints and relationships among entities
 - Supporting transaction management for admissions and discharges
 - Ensuring data reliability, security, and scalability
 - Efficient querying and reporting for decision-making
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8. Conclusion

The Emergency Hospital Resource & Bed Management System provides a structured and reliable approach to managing hospital resources during high-pressure emergency scenarios. By leveraging database concepts such as entity modeling, integrity constraints, transactions, and concurrency control, the system aims to improve operational efficiency and patient outcomes.

This project forms a strong foundation for implementing advanced DBMS concepts in subsequent phases, including ER modeling, relational schema design, SQL querying, triggers, and transaction handling.

9 Queries Supported by the System

The system supports the following categories of database queries:

A. Patient & Admission Queries

- Retrieve patient details using **patient ID** or **emergency severity level**
- List all **currently admitted patients** in a hospital or specific ward
- Identify patients assigned to a **particular doctor**
- Track **patient admission history** and **discharge status**

B. Bed & Ward Management Queries

- Identify **occupied and available beds** in each ward
- Locate **suitable beds** for incoming critical or serious patients

C. Resource & Inventory Queries

- Display current inventory levels of **critical medical resources** (e.g., oxygen cylinders, ventilators)

- Identify resources that fall **below predefined threshold levels**
- Generate **alerts for resource shortages**

D. Doctor & Staff Queries

- Retrieve **doctor availability** and area of specialization
- List patients assigned to a **specific doctor**

E. Administrative & Monitoring Queries

- Generate **hospital-wise summaries** of bed occupancy
 - Produce **emergency readiness reports** for health authorities
 - View **daily admission and discharge statistics**
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10. Users of the Database Application & Their Purposes

The database application is accessed by multiple stakeholders, each with defined responsibilities and controlled access permissions.

10.1 Hospital Administrators

- **Monitor overall bed occupancy and resource availability**
- **Manage hospital wards, beds, and inventory**
- **Analyze emergency preparedness and generate administrative reports**

10.2 Doctors

- **View assigned patients and their admission details**
- **Check availability of ICU and emergency beds**

- **Coordinate patient treatment during emergency situations**

10.3 Nurses & Emergency Ward Staff

- **Admit and discharge patients**
- **Assign beds based on patient condition**
- **Update real-time patient and resource status**

10.4 Health Authorities

- **Monitor hospital readiness during large-scale emergencies**
 - **Analyze availability of critical medical resources**
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11. Contribution

With combined discussions and meets, we all contributed equally in completing the deadline 1.
